

GATI SHAKTI VISHWAVIDYALAYA

C Programming Language

Lab Assignment – 2

Instructions:

1. Write all the program and execute. Maintain soft copy of all codes and prepare a Lab Record consisting hand written programs (10 Programs out of this list).
 2. Draw flow chart for any 5 programs.
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1. Write a program to swap the values of two variables with the help of third variable and without using third variable.
2. Write a program to show the use of sizeof() operator on 6 different data types.
3. Write a program to illustrate the use of AND and Left shift bitwise operator.
4. Write a program to take as input the radius of a circle and calculate and display its area and perimeter.
5. Write a program to determine largest of three number. (using If else)
6. Ram's basic salary is input through the keyboard. His dearness allowance is 40% of basic salary, and house rent allowance is 20% of basic salary. Write a program to calculate his gross salary.
7. Write a program to read a grade and display the equivalent description. (S = Excellent, A= Very Good, B = Good, C = Fair, D = Average, F = Fail)
8. Write a program to determine whether a character entered through keyboard is a special symbol or not.

Characters ASCII Values

A – Z 65 - 90

a – z 97 – 122

0 – 9 48 – 57

Special symbols 0 - 47, 58 - 64, 91 - 96, 123 – 127

9. Write a program to find a number which is divisible by 5 and 8 and not by 6 and 9.
10. Write a program to check whether a triangle is valid or not, when the three angles of the triangle are entered through keyboard. A triangle is valid if sum of all its angle is 180 degrees.
11. If the three sides of a triangle are entered through the keyboard, write a program to check whether the triangle is isosceles, equilateral, scalene or right angled triangle.
12. Write a program to read marks of student and display the equivalent grade. (using switch statement)
Marks ≥ 90 , A
80 $\leq$ Marks < 90 , B
70 $\leq$ Marks < 80 , C
60 $\leq$ Marks < 70 , D

50 <= Marks < 40, E

Marks <= 40, F

13. Write a program to check whether a given number is palindrome or not.

14. Write a program to check whether a given number is prime or not.

15. Write a program to produce the following pattern where number of line entered by keyboard:

a)

```
*  
**  
***  
****
```

b)

```
  *  
  **  
 ***  
****
```

c)

```
*****  
***  
***  
*
```

d)

```
*****  
  ***  
   **  
    *
```